SESSION 7 REFERENCE GUIDE: RELATIONAL WRANGLING, MERGING & RESHAPING

This guide provides a reference for combining separate tables using `pd.merge()`, stacking datasets with `pd.concat()`, and reshaping with `pd.pivot_table()`.

1. Relational Joins (`pd.merge()`)

Combines two DataFrames based on a shared key column.

```python

import pandas as pd

# Basic Inner Join (Intersection of keys)
inner_df = pd.merge(df_left, df_right, on="common_id", how="inner")

# Left Join (All rows from left table, matching from right)
left_df = pd.merge(df_left, df_right, on="key_col", how="left")

# Merging on differently named key columns
merged_df = pd.merge(
    df_orders,
    df_customers,
    left_on="cust_id",
    right_on="customer_id",
    how="left"
)
```

2. Table Concatenation (pd.concat())

Stacks DataFrames vertically or horizontally when columns match.

```python

# Vertical stacking (adding rows)
combined_rows = pd.concat([df_q1, df_q2], ignore_index=True)

# Horizontal stacking (adding columns)
combined_cols = pd.concat([df_part1, df_part2], axis=1)
```

3. Reshaping with Pivot Tables (`pd.pivot_table()`)

Summarizes data across two categorical dimensions into an executive matrix.

```python
pivot_summary = pd.pivot_table(
    df,
    values="order_revenue",       # Metric to aggregate
    index="customer_tier",        # Row categories
    columns="campaign_name",      # Column categories
    aggfunc="mean"                # Aggregation function (sum, mean, count)
)
```

4. The Relational Wrangling Workflow

```
[Inspect Tables] --> [Identify Common Keys] --> [Select Join Type
(Left/Inner)] --> [Execute pd.merge()] --> [Validate NaNs & Aggregate]
```